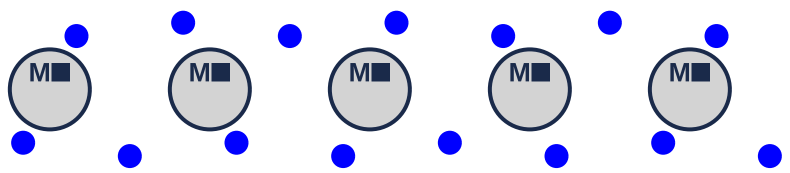
# Unit 5: Chemical Bonding

## Complete Study Notes – Grade 9 Chemistry

**Metallic Bonding – Sea of Electrons**




Delocalized electrons (blue) move freely among positive metal ions

# Introduction

Most of the elements in nature exist in combined states because atoms of some elements are unstable on their own. Except for a few elements (noble gases), atoms have an inherent tendency to combine and form molecules or compounds. Within molecules, atoms are held together by attractive forces.

A **chemical bond** is the force that holds atoms together to form molecules or compounds. Only **valence electrons** are involved in chemical bonding.

# 5.1 Chemical Bonding

## Why do atoms form chemical bonds?

Atoms form chemical bonds to achieve a more stable electron configuration, usually that of a noble gas (8 valence electrons).

## Octet Rule:

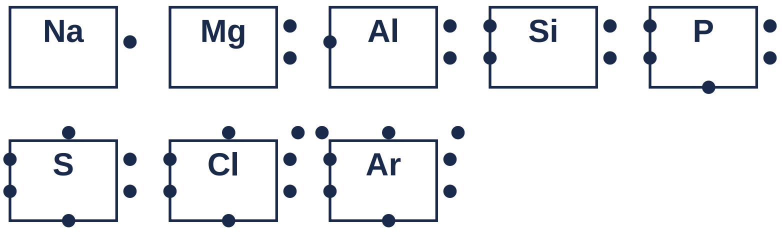
The **octet rule** states that atoms tend to form compounds in ways that give them eight valence electrons, thus achieving the electron configuration of a noble gas.

**Exceptions:** Hydrogen and helium need only 2 electrons (duet rule). Some elements can have expanded octets.

**Three ways atoms can satisfy the octet rule:**

1. **Losing valence electrons** – Metals (e.g., Na → Na■ + e■)
2. **Gaining valence electrons** – Nonmetals (e.g., Cl + e■ → Cl■)
3. **Sharing valence electrons** – Between nonmetals (e.g., H■, O■, N■)

**Lewis Symbols (Third Period Elements)**




# 5.2 Ionic Bonding

## 5.2.1 Formation of Ionic Bonding

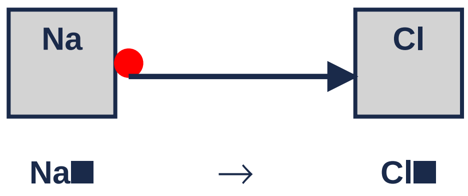
An **ion** is any atom or molecule with a net charge (positive or negative).

- **Cation** – positively charged ion (formed by losing electrons).
- **Anion** – negatively charged ion (formed by gaining electrons).

**Net electric charge = number of protons – number of electrons**

An **ionic bond** is a bond formed by the electrostatic attraction between two oppositely charged ions. Ionic bonds typically form between metals (which lose electrons) and nonmetals (which gain electrons).

**Ionic Bond Formation: NaCl**

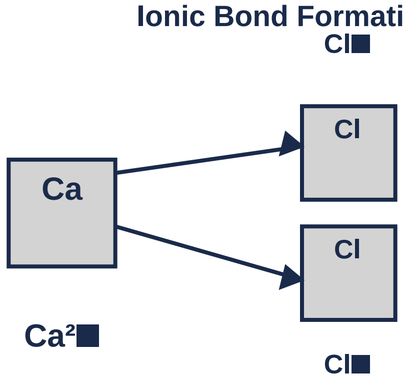



Na loses 1 electron → Na■; Cl gains 1 electron → Cl■

**Example – Sodium Chloride (NaCl):**

Na (2,8,1) loses 1 electron → Na■ (2,8). Cl (2,8,7) gains 1 electron → Cl■ (2,8,8). The electrostatic attraction between Na■ and Cl■ forms NaCl.

**Ionic Bond Formation: CaCl■**




**Example – Calcium Chloride (CaCl■):**

Ca (2,8,8,2) loses 2 electrons → Ca$^{2}$■ (2,8,8). Two Cl atoms each gain 1 electron → 2Cl■. Ca$^{2}$■ + 2Cl■ → CaCl■.

## 5.2.2 Lewis Formulas of Ionic Compounds

In the **Lewis symbol** for an atom, the chemical symbol is surrounded by dots representing valence electrons. These dots are arranged around the symbol with no more than two on each side.

Examples: Na· , ·Mg· , ·Al· , ·Si· , ·P· , ·S· , ·Cl· , ·Ar·

**Lewis structure for ionic compounds:**

- Na■ + :Cl■ → Na■[:Cl]■
- Ca$^{2}$■ + 2[:F]■ → Ca$^{2}$■[:F]■■

## 5.2.3 General Properties of Ionic Compounds

1. **Form Crystals:** Ionic compounds exist as crystalline solids with a regular, repeating three-dimensional structure.
2. **High Melting and Boiling Points:** Strong electrostatic forces require large amounts of energy to break. Example: NaCl melts at 801°C.
3. **Hard and Brittle:** The crystal structure is hard but shatters when struck due to repulsion of like charges.
4. **Soluble in Polar Solvents:** 'Like dissolves like' – ionic compounds dissolve in water (polar) but not in nonpolar solvents.
5. **Conduct Electricity:** In molten state or dissolved in water, ions are free to move and conduct electricity.
6. **High Density:** Ions are packed closely together due to strong electrostatic forces.

# 5.3 Covalent Bonding

## 5.3.1 Formation of Covalent Bond

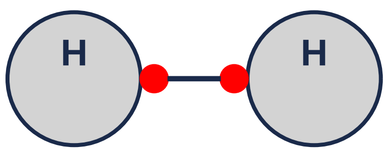
A **covalent bond** is formed when two atoms share one or more pairs of electrons. The shared electrons are counted as part of the valence shell of both atoms.

Covalent bonds occur between atoms with similar electronegativities (typically nonmetals).

**Types of covalent bonds:**

- **Single bond** – one pair of electrons shared (e.g., H■, Cl■).
- **Double bond** – two pairs of electrons shared (e.g., O■).
- **Triple bond** – three pairs of electrons shared (e.g., N■).

**Covalent Bond: H■**

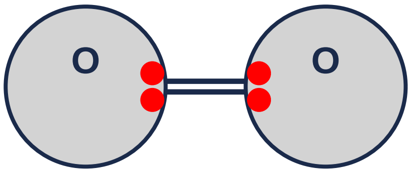



Each H shares 1 electron → single bond (H–H)

**Single Bond – Hydrogen (H■):**

Each H atom has 1 valence electron. They share one pair of electrons, each achieving a helium configuration (duet). H· + ·H → H:H (or H–H).

**Covalent Bond: O■**

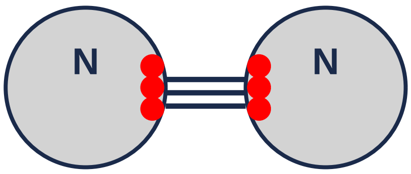



Each O shares 2 electrons → double bond (O=O)

**Double Bond – Oxygen (O■):**

Each O atom has 6 valence electrons. They share two pairs of electrons, each achieving an octet. O::O (or O=O).

**Covalent Bond: N■**




Each N shares 3 electrons → triple bond (N≡N)

**Triple Bond – Nitrogen (N■):**

Each N atom has 5 valence electrons. They share three pairs of electrons, each achieving an octet. N:::N (or N≡N).

## 5.3.2 Lewis's Formula of Covalent Molecules

The Lewis structure shows the arrangement of atoms and shared electron pairs.

- **H■O:** H–O–H (2 lone pairs on O)
- **NH■:** H–N–H with 1 lone pair on N (3 bonds, 1 lone pair)
- **CH■:** H–C–H with 4 bonds (No lone pairs on C)
- **CO■:** O=C=O (2 double bonds)

## 5.3.3 Polarity in Covalent Molecules

**Nonpolar covalent bond:** Electrons are shared equally between two identical atoms (e.g., H■, O■, Cl■).

**Polar covalent bond:** Electrons are shared unequally because of differences in electronegativity (e.g., HCl, H■O, NH■). One end has a partial positive charge (δ■) and the other has a partial negative charge (δ■).

Nonpolar vs Polar Covalent Bonds


**Guidelines for bond type:**

- Electronegativity difference 0–0.4: Nonpolar covalent.
- Electronegativity difference 0.5–2.0: Polar covalent.
- Electronegativity difference > 2.0: Ionic.

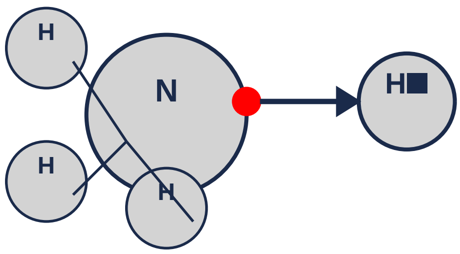
## 5.3.4 Coordinate Covalent Bond (Dative Bond)

A **coordinate covalent bond** is a covalent bond in which both shared electrons are provided by only one of the bonded atoms.

**Conditions for coordinate covalent bond:**

1. One atom must have a **lone pair** of electrons (e.g., NH■, H■O).
2. The other atom must have an **empty orbital** to accept the electrons (e.g., BF■, H■).

Coordinate Covalent Bond: NH■■




N donates lone pair to H■ → dative bond

**Examples:**

- **NH■■ (Ammonium ion):** NH■ + H■ → NH■■ (N donates lone pair to H■).
- **H■O■ (Hydronium ion):** H■O + H■ → H■O■ (O donates lone pair to H■).
- **BF■·NH■:** NH■ donates lone pair to BF■ (where B has only 6 valence electrons).

## 5.3.5 General Properties of Covalent Compounds

**1. Low Melting and Boiling Points:** Weak intermolecular forces (van der Waals) mean low melting/boiling points. Exception: diamond has very high melting point.

**2. Poor Conductors of Electricity:** Most covalent compounds do not conduct electricity (no free ions or electrons). Exception: graphite conducts electricity.

**3. Low Density:** Usually gases, liquids, or soft solids at room temperature.

4. **Solubility:** Dissolve in nonpolar solvents ('like dissolves like').

5. **Slow Reactions:** Reactions between covalent compounds are often slower than ionic reactions.

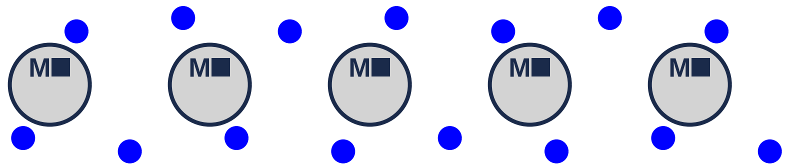
# 5.4 Metallic Bonding

## 5.4.1 Formation of Metallic Bond

A **metallic bond** is the attraction between the positive metal ions (cations) and the **delocalized electrons** (the 'sea of electrons') that are free to move throughout the metal structure.

In a metal, the valence electrons are not held by any particular atom but are free to move throughout the lattice. This model explains the properties of metals.

**Metallic Bonding – Sea of Electrons**




Delocalized electrons (blue) move freely among positive metal ions

## 5.4.2 Properties of Metallic Bond

1. **Electrical Conductivity:** Delocalized electrons can move and carry electric charge.

2. **Thermal Conductivity:** Electrons transfer kinetic energy rapidly, making metals good conductors of heat.

3. **Malleability and Ductility:** Layers of atoms can slide over each other without breaking the metallic bond.

4. **High Melting and Boiling Points:** Strong attraction between ions and delocalized electrons requires much energy to break.

5. **Metallic Luster:** Delocalized electrons reflect light.

**Factors affecting metallic bond strength:**

- **Number of delocalized electrons:** More electrons → stronger bond.
- **Charge of the metal ion:** Higher charge → stronger bond.
- **Size of the metal ion:** Smaller ion → stronger bond.

# Unit Summary

- A **chemical bond** is the force that holds atoms together. Only **valence electrons** are involved.
- The **octet rule** states that atoms tend to achieve 8 valence electrons (noble gas configuration).
- **Ionic bonds** form between metals and nonmetals through transfer of electrons. Ionic compounds have high melting points, are brittle, and conduct electricity when molten or

dissolved.

• **Covalent bonds** form between nonmetals through sharing of electrons. Types: single, double, triple.

• **Lewis structures** show valence electrons as dots. Used to represent both ionic and covalent compounds.

• **Polar covalent bonds** result from unequal sharing of electrons due to electronegativity differences.

• **Nonpolar covalent bonds** result from equal sharing of electrons.

• **Coordinate covalent bonds** form when one atom provides both shared electrons.

• **Metallic bonds** involve a 'sea of delocalized electrons' holding positive metal ions together.

• Metals are good conductors, malleable, ductile, and have high melting points due to metallic bonding.

## Key Terms

**Anion –** a negatively charged ion (formed by gaining electrons).

**Cation –** a positively charged ion (formed by losing electrons).

**Chemical bond –** the force that holds atoms together.

**Coordinate covalent bond –** a bond where both shared electrons come from one atom.

**Covalent bond –** a bond formed by sharing electrons.

**Double bond –** a covalent bond with two shared pairs of electrons.

**Ionic bond –** a bond formed by electrostatic attraction between oppositely charged ions.

**Lewis structure –** a representation showing valence electrons as dots.

**Metallic bond –** the attraction between positive metal ions and delocalized electrons.

**Octet rule –** atoms tend to gain, lose, or share electrons to achieve 8 valence electrons.

**Polar molecule –** a molecule with unequal distribution of charge.

**Single bond –** a covalent bond with one shared pair of electrons.

**Triple bond –** a covalent bond with three shared pairs of electrons.

**Valence electrons –** electrons in the outermost shell of an atom.

**End of Unit 5 – Chemical Bonding**